CONCEPT 2 — MULTIGENERATIONAL HOUSE

Two storeys · drive-through west carport · central skylight stair + reserved lift · ground-floor suite for the older couple



Blender render — street view (golden hour)

A two-storey multigenerational house for three generations. A covered drive-through carport and service bay runs down the west side, open to the road at the front and to the garden at the back. The entrance hall (Huyền Quan) opens to a central stair under a skylight, with a 1.6 × 1.6 m shaft beside it reserved for a future lift. The ground floor works on its own for the older couple: a master suite, an accessible bathroom and a flexible room that can become a bedroom or study. Living and dining face the north garden through large openings, with the kitchen and a compact wet kitchen beside them. Upstairs there are three bedrooms for the daughter's family, a family lounge and a shaded north balcony. The east wall is blank against the assumed 3-storey neighbour. Nothing is built on the DNN land except the boundary walls down to the water.



Concept 2 brief image (inspiration)

THE CONCEPT IN NUMBERS

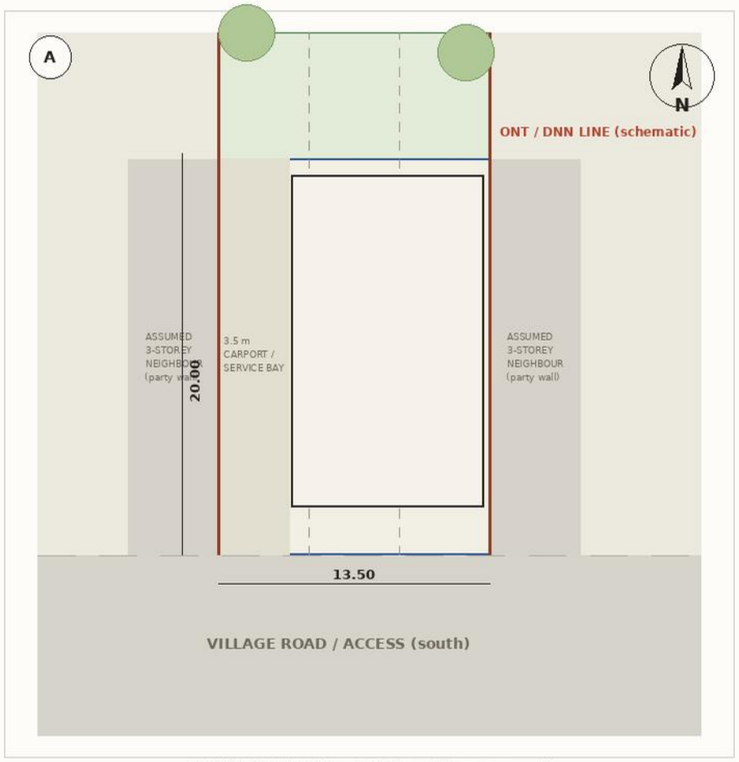
Footprint	158.4 m² (60% of ONT band)
Ground floor	158 m² enclosed
Upper floor	137 m² enclosed + 21 m² balcony
Total enclosed	296 m²
Constraint check	PASS — all structures in ONT; DNN planting only

Walkthrough test: 21/21 rooms reachable

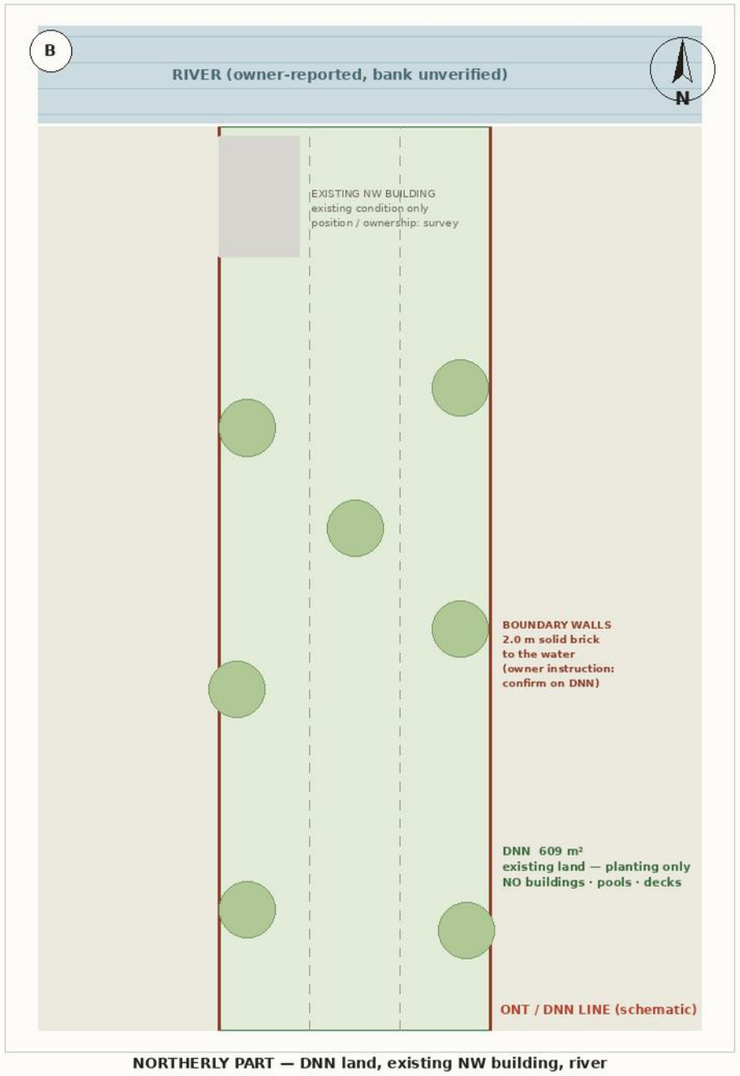
The brief asks for a 3.3–3.6 m drive-through carport plus a full house inside the 13.5 m ONT frontage. That leaves a 9.6 m wide house, kept back 0.3 m from the east party wall so the eaves stay on the plot. The future lift shaft (1.6 × 1.6 m) is reserved on both floors next to the stair, so a lift can go in later without structural changes. The carport is open at both ends; driving out to the garden crosses DNN land as access only, with no paving or structure there.

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

SITE PLAN · 1:200 at A3 · shown in two parts so the narrow 13.5 m plot stays readable



SOUTHERLY PART — ONT band, house, road



NORTHERLY PART — DNN land, existing NW building, river

0 10 20 30 40 50 metres (assumed — no survey)

NOTES

Schematic site from owner information only: three side-by-side 4.5 m titles, about 13.5 × 65 m. No survey. Corners, levels, bank edge and road reserve are unverified.

Structures are shown ONLY inside the ONT road-side band (3 × 88.8 = 266.4 m²). The 609 m² DNN band is existing land, shown as planting only.

The title sketch shows 6.5 m to the road centreline. It is NOT treated as a verified setback.

Assumed: a 3-storey house on each side built to the property line to about the same 20 m depth. So the east wall is blank (party wall) and the west wall has high-level windows only.

Side boundary walls run the full length to the water on both property lines (owner instruction). This is the only structure on DNN land. Confirm it is permitted.

The existing north-west building is an existing condition only. Its position and ownership are to be surveyed. It is not altered.

Subject to legal merger of the three ONT sections, setbacks, river corridor and permits.

KEY

- ONT — buildable band
- DNN — land / planting only
- service passage
- existing NW building
- assumed 3-storey neighbours
- boundary walls
- house footprint
- assumed / context

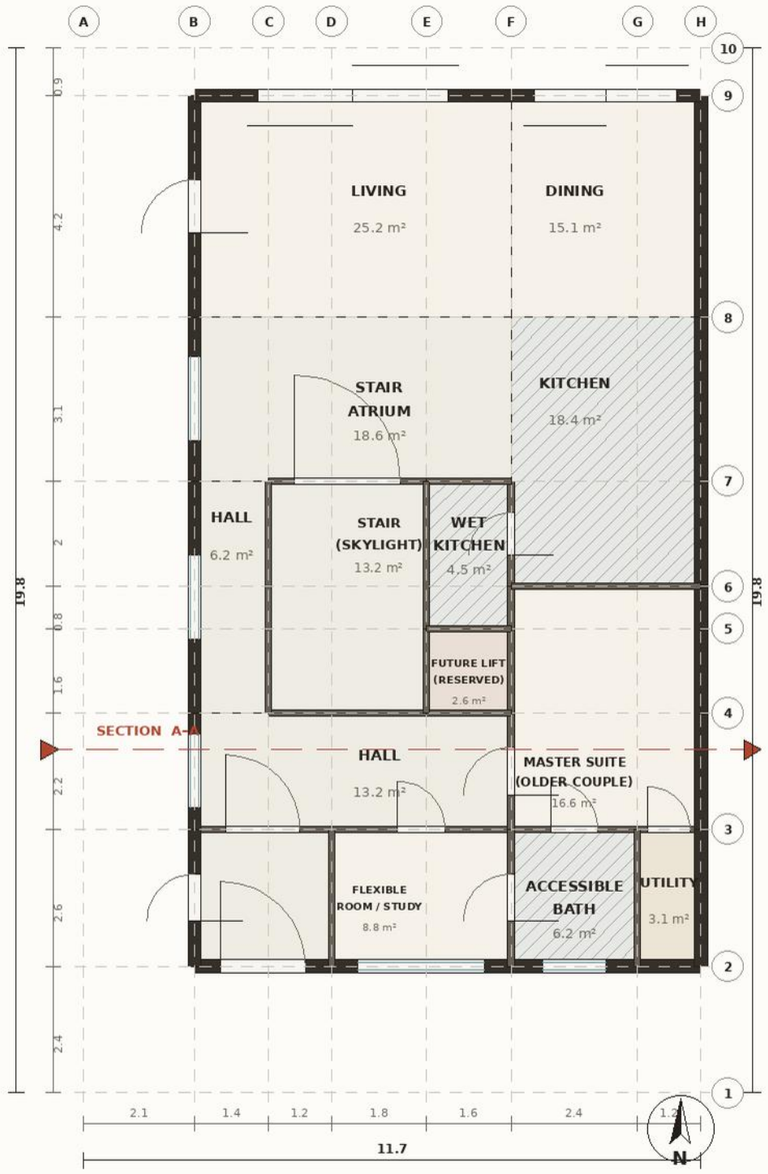
CONCEPT 2 — MULTIGENERATIONAL HOUSE

Two storeys · drive-through west carport · central skylight stair + reserved lift · ground-floor suite for the older couple

SCALE	SHEET	STATUS
1:200 at A3	C2-01 site plan	FOR REVIEW *not for construction*

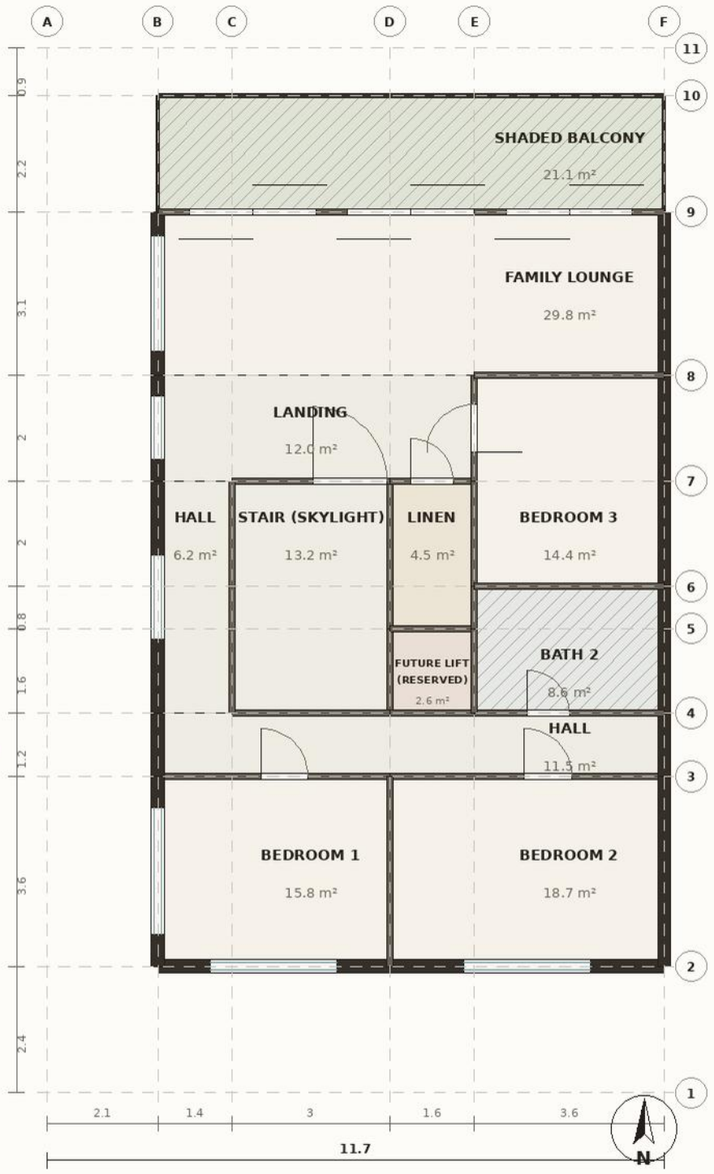
enclosed 158.4 m² · level 1 of 2

GROUND FLOOR · FFL +0.45 / +3.75



enclosed 137.3 m² · level 2 of 2

UPPER FLOOR · FFL +3.75 / +6.85



0 10 20 30 40 50 metres (assumed — no survey)

KEY

- HABITABLE
- SERVICE / WET
- CIRCULATION
- STORAGE
- SPECIAL
- OUTDOOR
- PLANT / VOID

NOTES

CONCEPT ONLY — NOT FOR CONSTRUCTION. Dimensions are taken from owner-supplied title information; no survey. Subject to survey, legal merger, setbacks and permits.

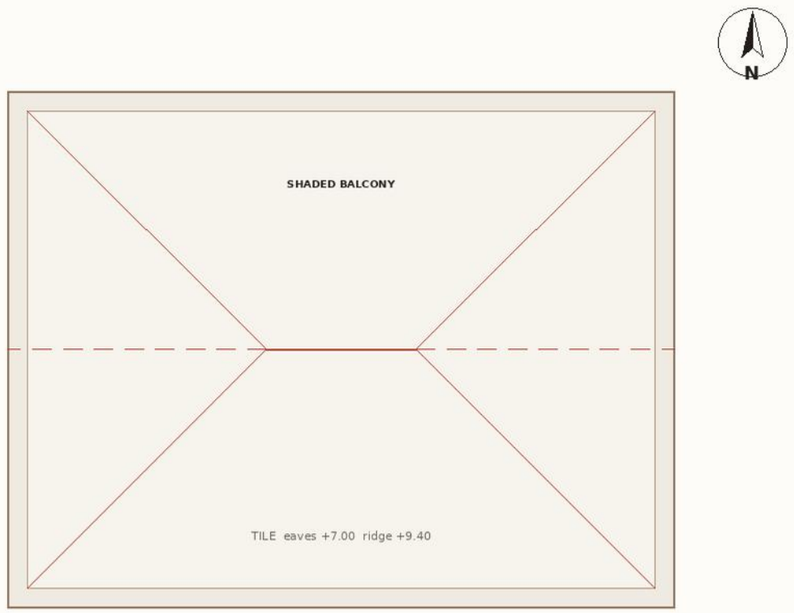
Areas are measured from the room rectangles; the balcony and light court are excluded from enclosed area.

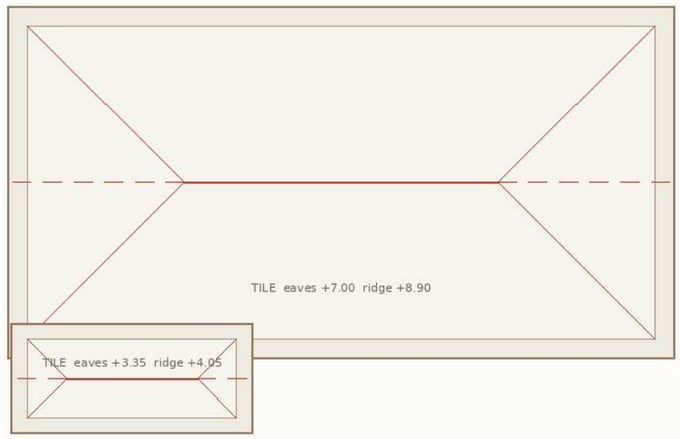
CONCEPT 2 — MULTIGENERATIONAL HOUSE

Two storeys · drive-through west carport · central skylight stair + reserved lift · ground-floor suite for the older couple

SCALE	SHEET	STATUS
1:100 at A3	C2-02	FOR REVIEW
	floor plans 1–2	not for construction

ROOF PLAN · looking down





0 10 20 30 40 50 metres (assumed — no survey)

KEY

- roof covering
- ridge / hip
- fascia / eaves
- open slot below

ROOF KEY

- concrete tile on timber — eaves +7.00, ridge +9.40
- ridge / hip line
- fascia / eaves edge, 1.05 m overhang
- rain-garden slot, open to the sky
- outline of the level below (hidden line)

NOTES

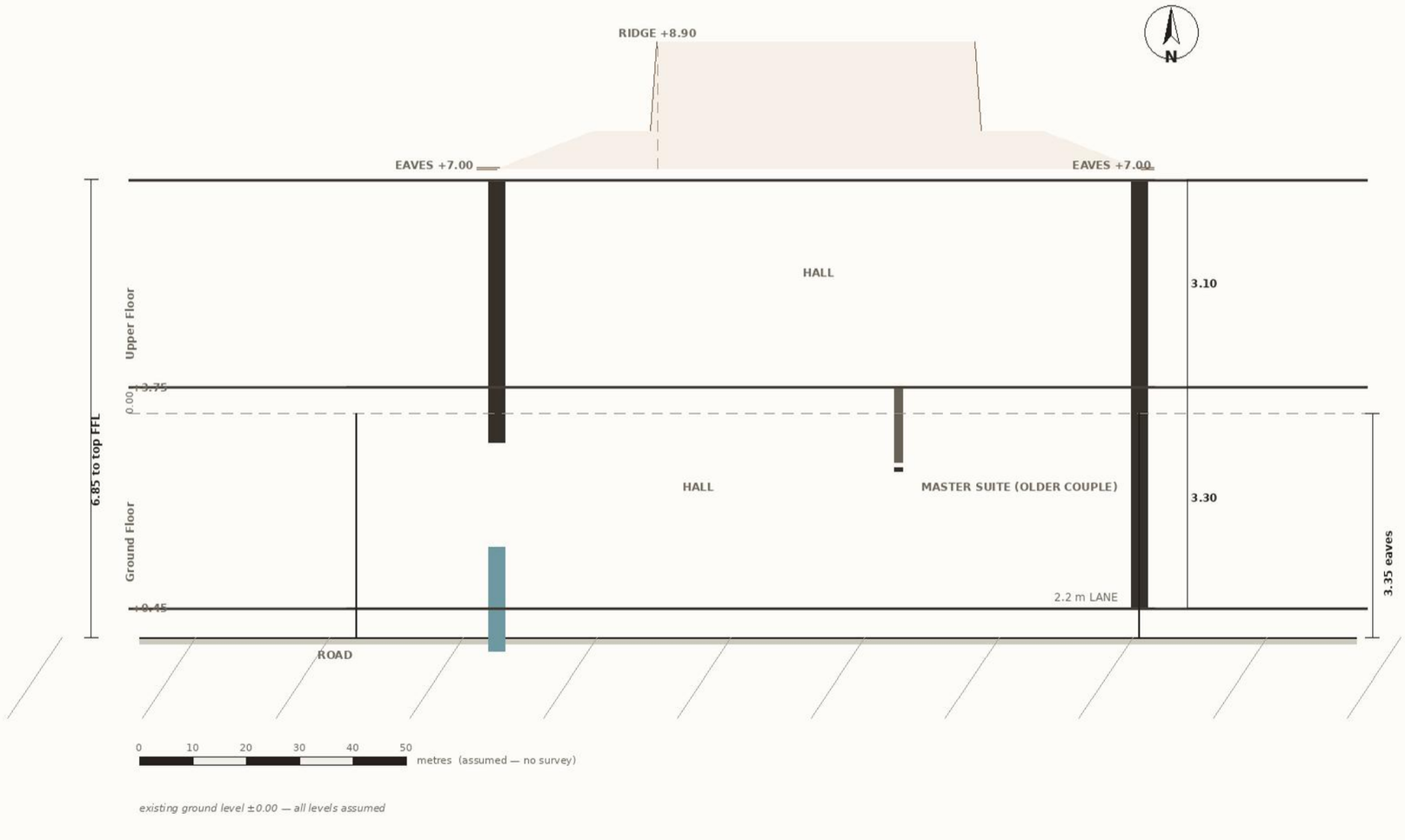
Ridge and eaves heights come from the option massing: eaves +7.00 m, ridge +9.40 m above datum, 0.30 m overhang all round.
Hips, ridges and the rain-garden slot are derived from the same rectangles as the plans and the 3D model.
Rainwater from the slot and the roof gardens is the only visible drainage; confirm the neighbour-boundary falls before detailing.

CONCEPT 2 — MULTIGENERATIONAL HOUSE

Two storeys · drive-through west carport · central skylight stair + reserved lift · ground-floor suite for the older couple

SCALE	SHEET	STATUS
1:100 at A3	**C2-03** roof plan	**FOR REVIEW** *not for construction*

SECTION A–A · looking north at y = 6.50 m (assumed cut)



existing ground level ±0.00 — all levels assumed

KEY

- cut external wall, 0.30
- cut internal wall, 0.15
- glazing in the cut
- roof build-up in section

NOTES

Datum 0.00 = existing road and ground level. Finished ground floor +0.45 m, first +3.75 m, eaves +3.35 m, ridge +9.40 m.

Slabs, walls and levels come from the same rectangles as the plans, so the section and the plans can never disagree.

The 2.2 m rain-garden slot is open to the sky on every level: draw it as a full-height void when the structure is detailed.

The cut is at y = 6.50 m. Move it on the plans to suit a later survey.

CONCEPT 2 — MULTIGENERATIONAL HOUSE

Two storeys · drive-through west carport · central skylight stair + reserved lift · ground-floor suite for the older couple

SCALE	SHEET	STATUS
1:100 at A3	**C2-04** section A–A	**FOR REVIEW** *not for construction*

AREA SCHEDULE · net internal, measured from the plan rectangles

Room	Type	Area m²	Cum. m²
Ground Floor · FFL +0.45			
Dining	Habitable	15.1	15.1
Flexible Room / Study	Habitable	8.8	24.0
Living	Habitable	25.2	49.2
Master Suite (older couple)	Habitable	16.6	65.7
Accessible Bath	Service / Wet	6.2	72.0
Stair Atrium	Circulation	18.6	90.6
Entry (Huyền Quan)	Circulation	6.8	97.3
Hall	Circulation	13.2	110.5
Hall	Circulation	6.2	116.7
Kitchen	Service / Wet	18.4	135.0
Future Lift (reserved)	Special	2.6	137.6
Stair (skylight)	Circulation	13.2	150.8
Utility	Storage	3.1	153.9
Wet Kitchen	Service / Wet	4.5	158.4
Level total, enclosed		**158.4**	
Upper Floor · FFL +3.75			
Bedroom 1	Habitable	15.8	15.8
Bedroom 2	Habitable	18.7	34.6
Bedroom 3	Habitable	14.4	49.0
Family Lounge	Habitable	29.8	78.7
Shaded Balcony	Outdoor	21.1	99.8
Bath 2	Service / Wet	8.6	108.5
Hall	Circulation	11.5	120.0
Hall	Circulation	6.2	126.2
Landing	Circulation	12.0	138.2
Future Lift (reserved)	Special	2.6	140.7
Linen	Storage	4.5	145.2
Stair (skylight)	Circulation	13.2	158.4
Level total, enclosed		**137.3**	

AREA SUMMARY	Area
Level 1 — Ground Floor (enclosed)	158.4 m²
Level 2 — Upper Floor (enclosed)	137.3 m²
Total enclosed floor area	**295.7 m²**

ORDER OF COST — TURNKEY, NORTHERN VIET NAM

INDICATIVE BUILD COST	M VND	+10% prov.
Basic build — 4.9 M VND/m²	1 449	1 594
Mid build — 6.5 M VND/m²	1 922	2 114
High build — 8.5 M VND/m²	2 513	2 765

NOTES

Areas are net internal floor areas computed from the room rectangles. Terraces, loggias, the rain-garden slot, the plant deck and roof gardens are excluded from the totals but shown on the plans.

The build-cost range is a rule of thumb from published Vietnamese cost guides for northern Viet Nam, not a quotation. It excludes site works, boundary treatment and furniture.

Confirm with a local quantity surveyor and at least two builders before committing to a budget.

NOTES

No habitable room here is below 2.7 m clear. The ground floor is a complete one-level home — bedroom, bathroom, living, dining and kitchen all at +0.45 — so the family can live on one level for as long as they want to.

Order-of-cost at 4.9–8.5 M VND/m² turnkey (northern Viet Nam) is in the table on the left. It is a rule of thumb, not a quotation.

ROOM TYPE KEY

- HABITABLE
- SERVICE / WET
- CIRCULATION
- STORAGE
- SPECIAL
- OUTDOOR
- PLANT / VOID

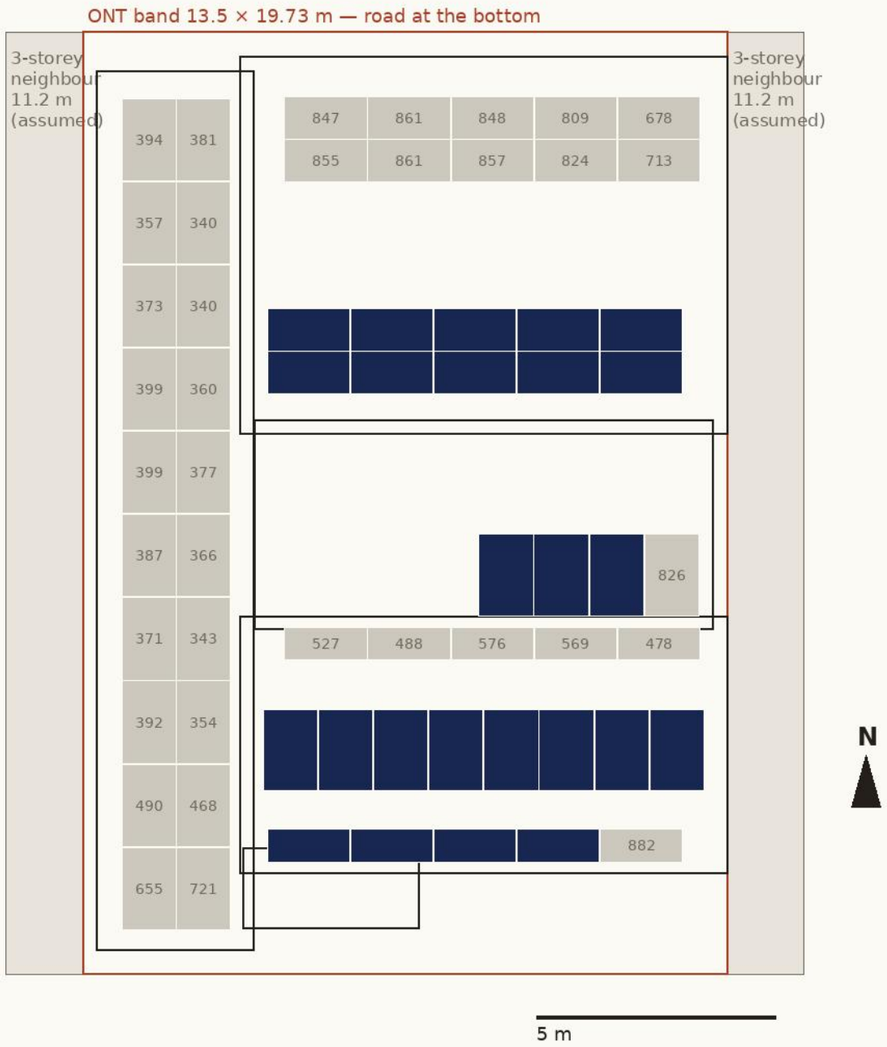
CONCEPT 2 — MULTIGENERATIONAL HOUSE

Two storeys · drive-through west carport · central skylight stair + reserved lift · ground-floor suite for the older couple

SCALE	SHEET	STATUS
as noted	**C2-05** area schedule	**FOR REVIEW** *not for construction*

Rooftop solar — energy per season

25 × 440 Wp panels = 11.0 kWp on the roof · latitude 21°N (rounded) · PVGIS 5.3 TMY (ERA5), typical year · shaded by the assumed 3-storey neighbours



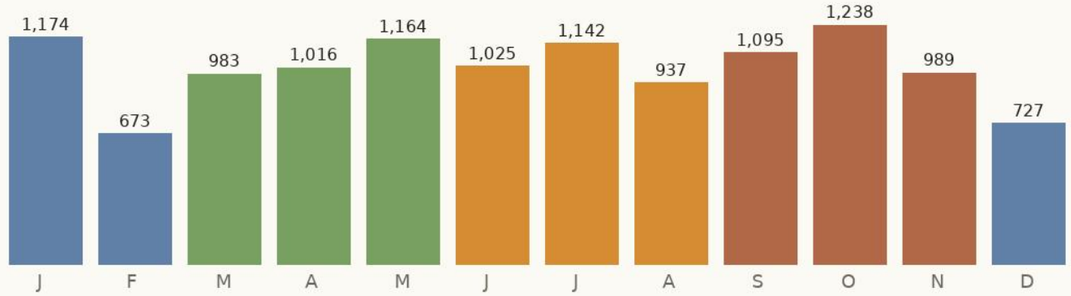
Recommended panels Fit but not worth it: number = kWh per kWp a year (an unshaded ideal panel gives 1281)

Roof surface	Tilt / faces	Fit / used	kWh/kWp	no neighbours
Front roof south slope	53° S	5 / 4	1015	1100
Front roof flat top	10° S	8 / 8	1199	1275
Front roof north slope	53° N	5 / 0	528	625
Middle roof flat deck	10° S	4 / 3	976	1192
Rear roof south slope	38° S	10 / 10	1079	1198
Rear roof north slope	38° N	10 / 0	815	884
Side lean-to roof	9° W	20 / 0	413	942

ENERGY FROM THE RECOMMENDED ARRAY

Season	Months	kWh	kWh / day	Share
Winter	Dec–Feb	2,574	28.6	21 %
Spring	Mar–May	3,163	34.4	26 %
Summer	Jun–Aug	3,104	33.7	26 %
Autumn	Sep–Nov	3,322	36.5	27 %
Year		**12,163**	**33.3**	

Without the assumed neighbours the same panels would give 13,258 kWh a year (+9 %). All 62 panels that physically fit would give 21,301 kWh, but the extra ones are on shaded or east/west/north slopes.



IS IT WORTH IT? (indicative)

	Recommended	Best-value size
Panels / kWp	25 / 11.0	6 / 2.6
Installed cost	154 M VND	37 M VND
Bill saving, first year	9 M VND	7 M VND
Output used in the house	21 %	60 %
Share of the bill-kWh covered	36 %	29 %
Simple payback	22 years	6 years
Net over 25 years	21 M VND	123 M VND

Assumes: an example air-conditioned family using about 6,890 kWh a year (bill 21 M VND a year), EVN household tiers from 1 Nov 2026 + 8 % VAT, 14 M VND per kWp installed, nothing paid for power sent to the grid (the buy-back rate is not yet published), 0.5 %/yr panel ageing, 1 %/yr upkeep. Surplus power is the limit: the house uses most of its power in the evening. Try other numbers at ali3b.github.io/home-design/solar.html.

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Batteries and living off-grid

Hour-by-hour over a typical year · battery 5 M VND per usable kWh incl. hybrid inverter, replaced once (year 13, 60 %) · nothing paid for surplus power

WITHOUT AIR-CON — example household using about 3,600 kWh a year (bill 9 M VND a year)

Panels	Battery	Cost	From sun: W Sp Su Au	Year	No-grid days	Saving / yr	Payback	Net over 25 years
6 (2.6 kWp)	none	37 M VND	28 34 36 32	32 %	0	4 M VND	12 yrs	39 M VND
6 (2.6 kWp)	**5 kWh**	**62 M VND**	58 73 74 71	**69 %**	**0**	**7 M VND**	**10 yrs**	**75 M VND ← best value**
6 (2.6 kWp)	10 kWh	87 M VND	66 88 88 87	82 %	174	8 M VND	12 yrs	59 M VND
6 (2.6 kWp)	15 kWh	112 M VND	66 89 90 89	84 %	196	8 M VND	22 yrs	21 M VND
25 (11.0 kWp)	none	154 M VND	34 41 44 38	40 %	0	4 M VND	> 25 yrs	-91 M VND
25 (11.0 kWp)	5 kWh	179 M VND	76 85 88 81	83 %	0	8 M VND	> 25 yrs	-46 M VND
25 (11.0 kWp)	10 kWh	204 M VND	98 100 100 100	99 %	358	9 M VND	> 25 yrs	-57 M VND
25 (11.0 kWp)	15 kWh	229 M VND	99 100 100 100	100 %	360	9 M VND	> 25 yrs	-97 M VND

WITH AIR-CON — example household using about 6,890 kWh a year (bill 21 M VND a year)

Panels	Battery	Cost	From sun: W Sp Su Au	Year	No-grid days	Saving / yr	Payback	Net over 25 years
6 (2.6 kWp)	none	37 M VND	27 31 27 29	29 %	0	7 M VND	6 yrs	123 M VND
6 (2.6 kWp)	**5 kWh**	**62 M VND**	51 49 34 48	**43 %**	**0**	**11 M VND**	**6 yrs**	**162 M VND ← best value**
6 (2.6 kWp)	10 kWh	87 M VND	57 52 34 51	46 %	1	11 M VND	8 yrs	131 M VND
6 (2.6 kWp)	15 kWh	112 M VND	57 52 34 51	46 %	1	11 M VND	11 yrs	91 M VND
25 (11.0 kWp)	none	154 M VND	34 38 37 35	36 %	0	9 M VND	22 yrs	21 M VND
25 (11.0 kWp)	5 kWh	179 M VND	69 62 52 60	59 %	0	14 M VND	16 yrs	95 M VND
25 (11.0 kWp)	10 kWh	204 M VND	96 84 66 84	79 %	123	18 M VND	15 yrs	141 M VND
25 (11.0 kWp)	15 kWh	229 M VND	97 95 79 95	89 %	201	19 M VND	16 yrs	144 M VND

CAN THE HOUSE LIVE OFF-GRID?

Without air-con: YES in a typical year with 25 panels (11.0 kWp) and a 40 kWh battery, about 354 M VND. That costs 296 M VND more over 25 years than staying on the grid.

Nearly: 25 panels with 10 kWh covers all but 21 kWh a year (7 short days, the worst needing 4.1 kWh).

With air-con: YES in a typical year with 62 panels (27.3 kWp, every panel that fits, including the shaded slopes) and a 40 kWh battery, about 582 M VND. That costs 301 M VND more over 25 years than staying on the grid.

Nearly: 62 panels with 20 kWh covers all but 20 kWh a year (11 short days, the worst needing 7.2 kWh).

A "typical year" leaves out the worst years. Northern winters bring weeks of drizzle and low cloud, so an off-grid house needs a margin: about 30 % more battery and panels, or a small generator for bad spells. Off-grid also wastes the summer surplus. Staying connected with a 5–10 kWh battery gets most of the benefit for far less money, and keeps the lights on in power cuts.

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Blender views

Cycles renders from the same data as the drawings and the walkthrough



Street view



From the river end, between the boundary walls

Blender views

Cycles renders from the same data as the drawings and the walkthrough



Drive-through carport / service bay



Stair atrium under the skylight

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Blender views

Cycles renders from the same data as the drawings and the walkthrough



Aerial

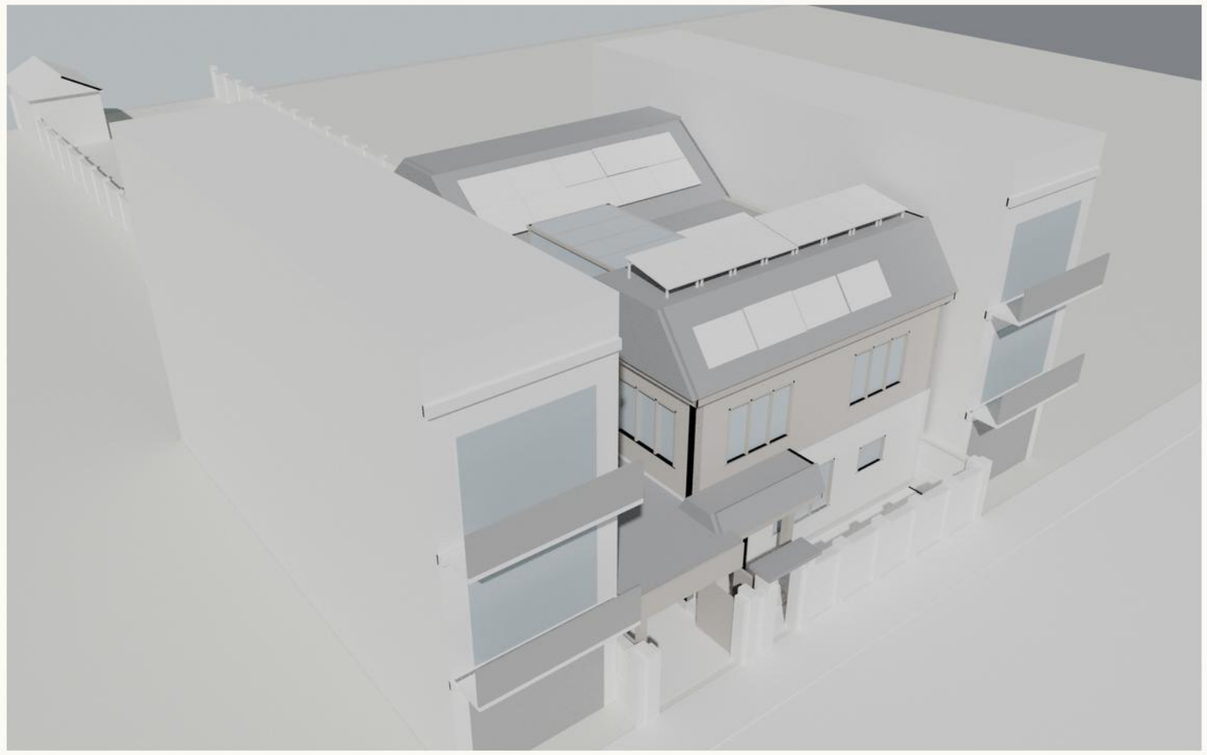


Upper balcony looking north

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Blender views

Cycles renders from the same data as the drawings and the walkthrough



Clay massing

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Walkthrough (Unity)

Interactive model: these are fixed viewpoints you can jump to with keys 1–9



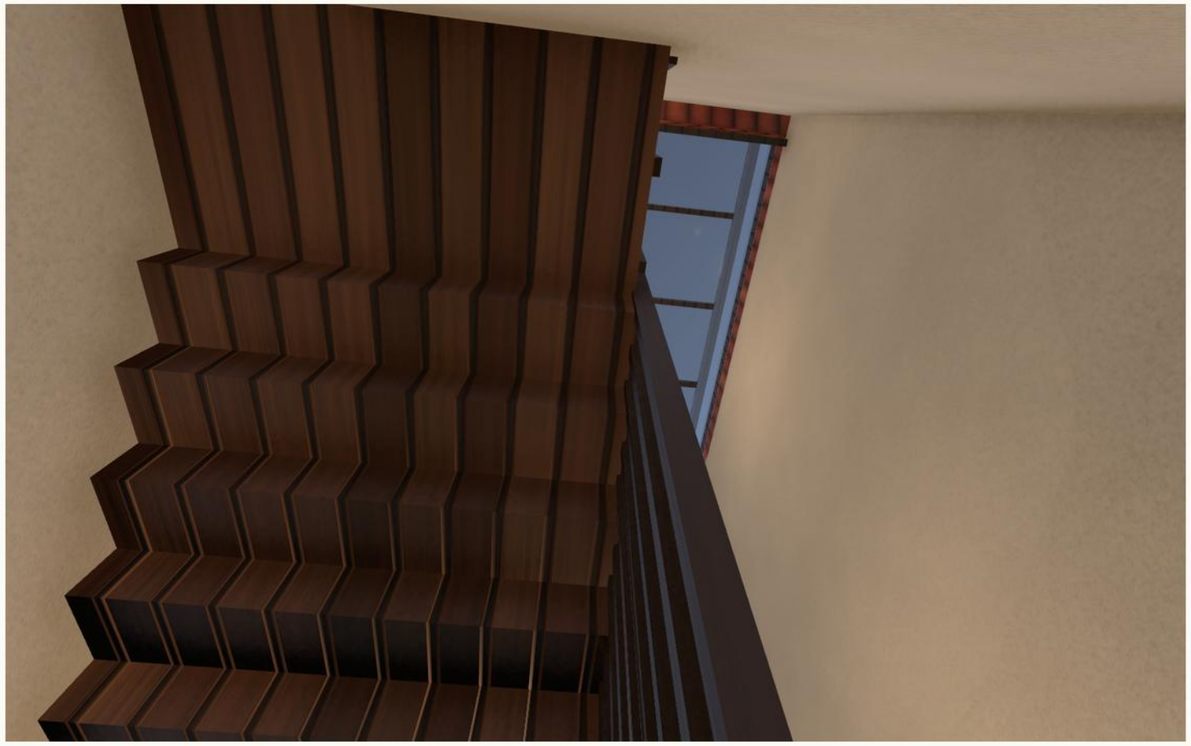
Street



Carport



Entry



Atrium

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Walkthrough (Unity)

Interactive model: these are fixed viewpoints you can jump to with keys 1–9



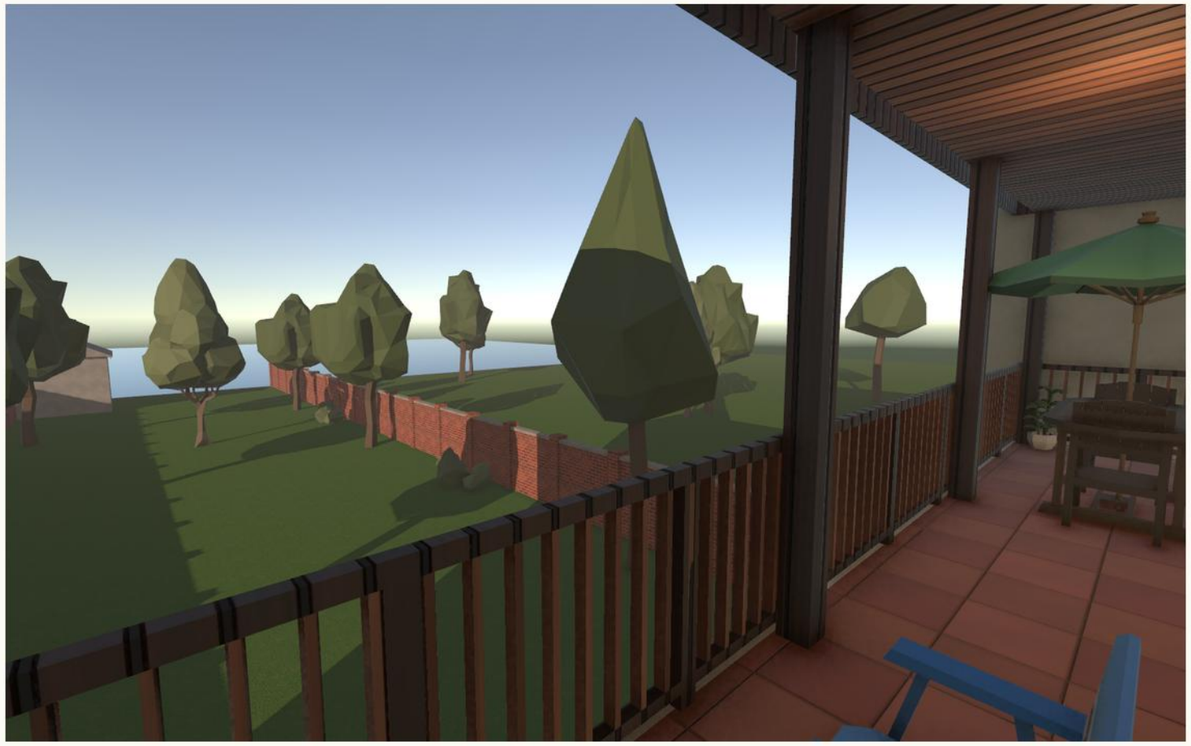
Living



Master



Lounge



Balcony

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Walkthrough (Unity)

Interactive model: these are fixed viewpoints you can jump to with keys 1–9



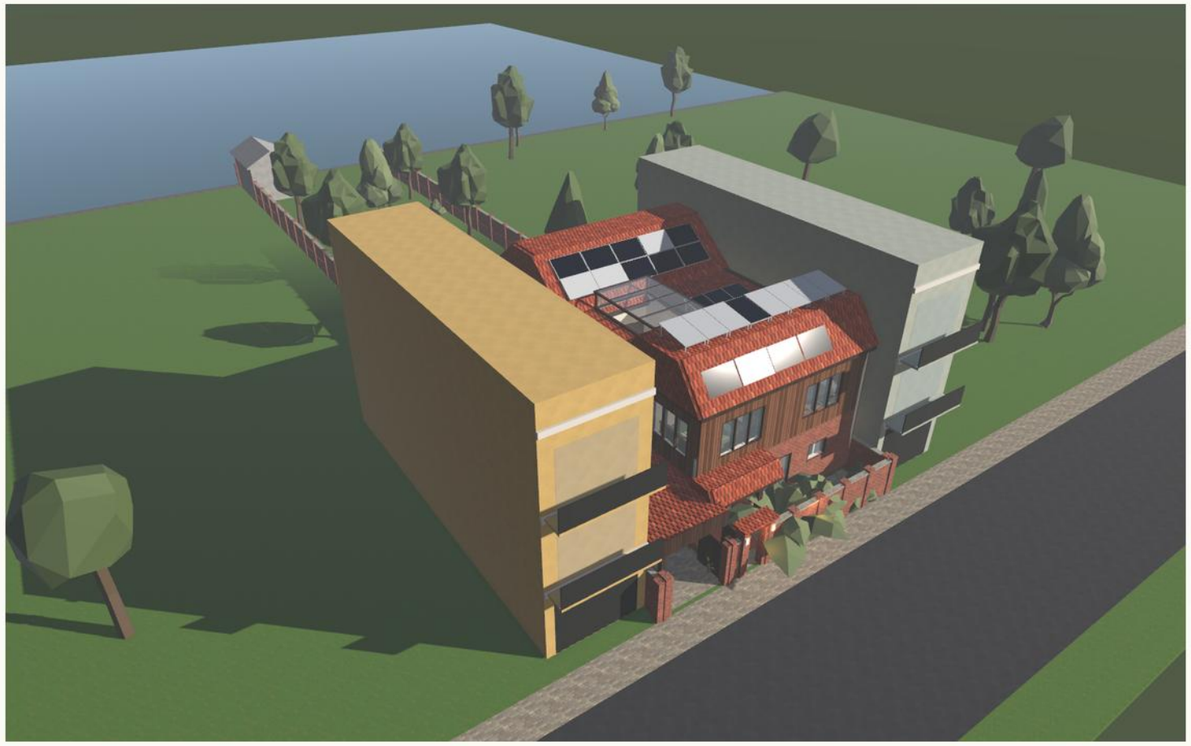
Aerial



Riverside



Dollhouse Ground

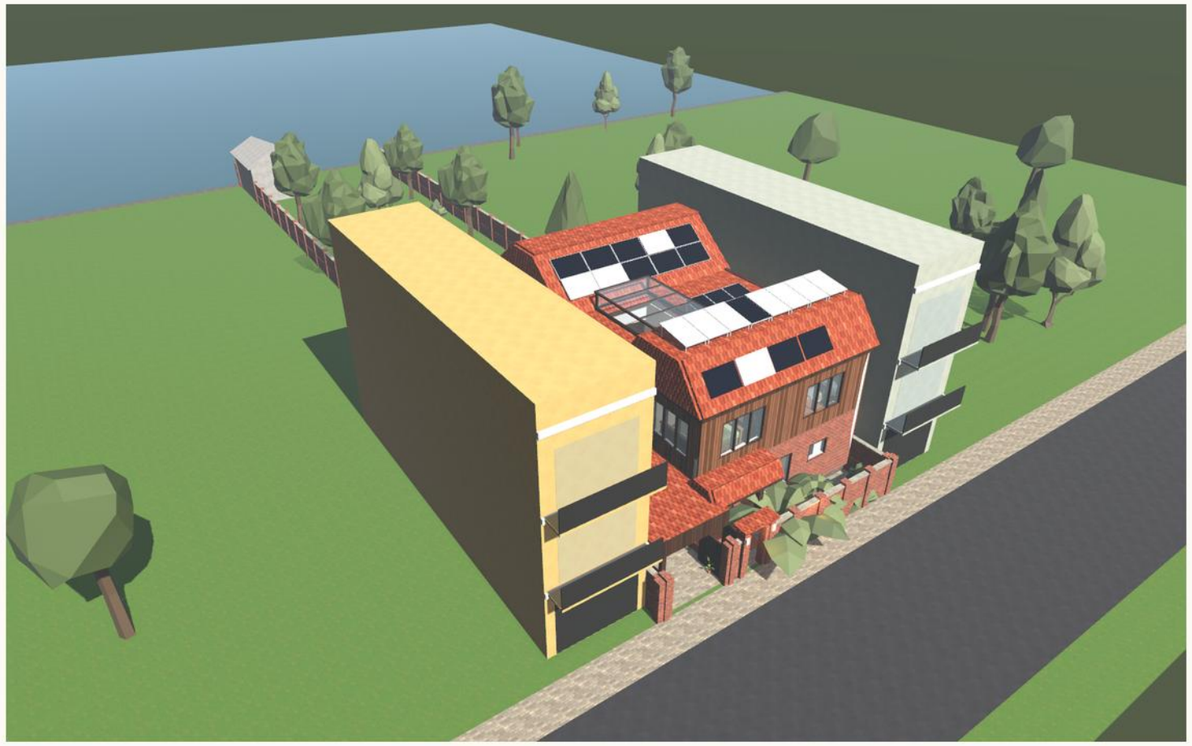


Sun study: winter solstice, 09:00 solar time

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Walkthrough (Unity)

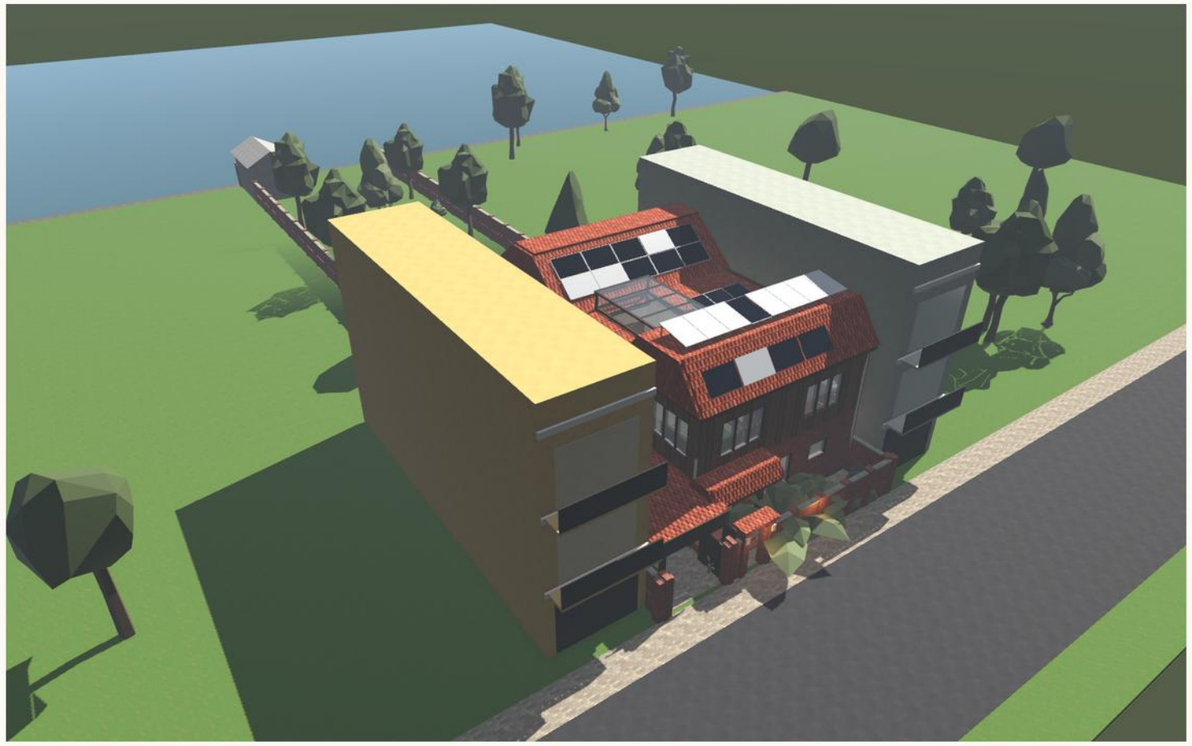
Interactive model: these are fixed viewpoints you can jump to with keys 1–9



Sun study: winter solstice, 12:00 solar time



Sun study: winter solstice, 15:00 solar time



Sun study: summer solstice, 09:00 solar time

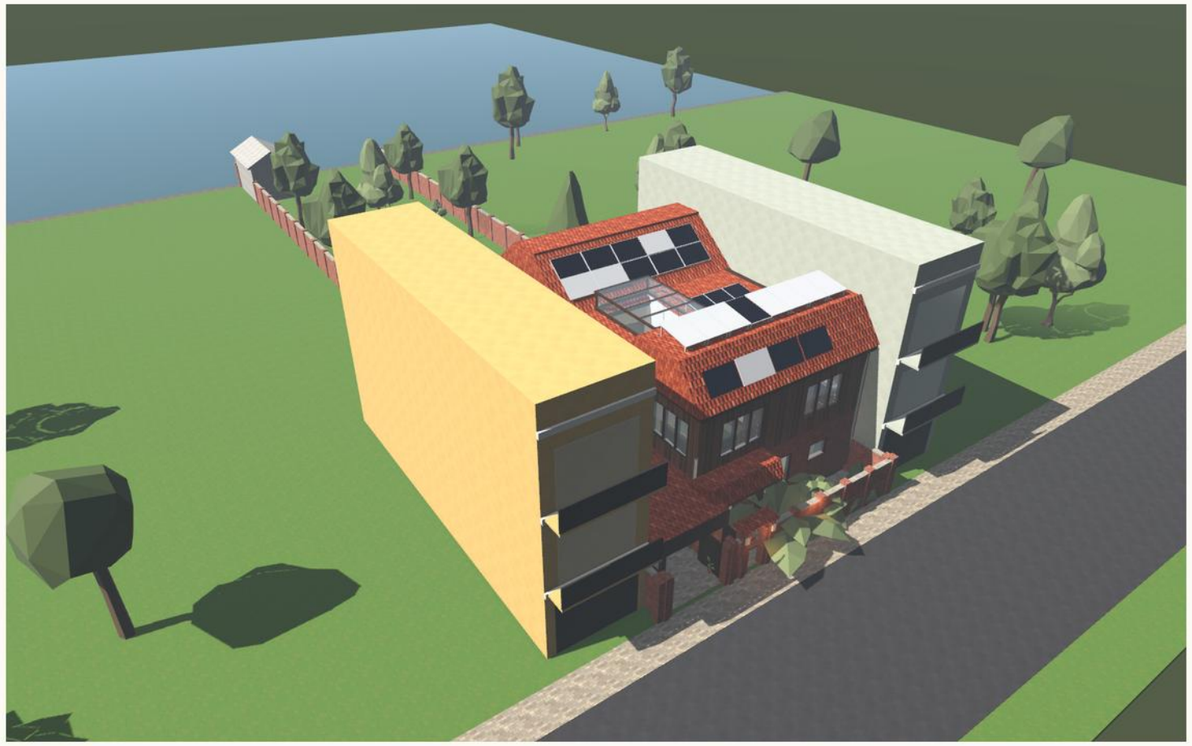


Sun study: summer solstice, 12:00 solar time

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits

Walkthrough (Unity)

Interactive model: these are fixed viewpoints you can jump to with keys 1–9



Sun study: summer solstice, 15:00 solar time

CONCEPT ONLY — NOT FOR CONSTRUCTION · subject to survey, legal merger, setbacks and permits